

Qubit Definition

Structure and physical meaning of a two-level quantum system

A **qubit** is the fundamental unit of quantum information.

- Mathematically, it is a quantum system whose state lives in a **two-dimensional Hilbert space**.
- Physically, it is any system that has **exactly two distinguishable states**.

examples include:

- spin $-\frac{1}{2}$ particles.
- two energy levels of an atom
- polarization of a photon
- superconducting qubits

In this chapter, we formalize what a qubit is, how its state is represented, and how measurements act on it.

State Space of a Qubit

A qubit is described by a normalized vector in a 2-dimensional Hilbert space:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The states $|0\rangle$ and $|1\rangle$ form the **computational basis**, which is the standard basis used in quantum information.

They are typically chosen as the eigenstates of σ_z :

$$\sigma_z |0\rangle = + |0\rangle; \quad \sigma_z |1\rangle = - |1\rangle.$$

This basis plays the same role as the classical bit values 0 and 1.

Global Phase and Physical Degrees of Freedom

A qubit state is defined up to a global phase:

$$|\psi\rangle \sim e^{i\gamma} |\psi\rangle.$$

This phase has no physical effect on any measurement outcome. Combined with normalization, this means that a **qubit has two real degrees of freedom**.

This is why the state of a qubit can be represented as a point on a sphere (the Bloch sphere), which will be introduced in the next chapter.

Canonical Parametrization of a Qubit

Because a qubit has only two physical degrees of freedom, we can parametrize its state using two angles θ and ϕ :

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle.$$

with:

$$0 \leq \theta \leq \pi \quad , \quad 0 \leq \phi \leq 2\pi.$$

Why this parametrization?

Start from the general state:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

Write the amplitudes in polar form:

$$\alpha = |\alpha| \quad , \quad \beta = e^{i\phi} |\beta|$$

Normalization gives:

$$|\alpha|^2 + |\beta|^2 = 1.$$

So we can set:

$$|\alpha| = \cos \frac{\theta}{2} \quad , \quad |\beta| = \sin \frac{\theta}{2}.$$

This parametrization:

- automatically enforces normalization
- isolates the relative phase $e^{i\phi}$
- uses exactly two real parameters $\rightarrow$ the physical degrees of freedom

This is the natural coordinate system for qubit states.

Measurement of a Qubit

Measurement is the process that extracts classical information from a quantum state. It is inherently probabilistic and **depends on the basis** in which the measurement is performed.

i. Measurement in the computational basis

Given:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

A measurement in the $\{|0\rangle, |1\rangle\}$, basis :

- outcome 0 with probability $|\alpha|^2$
- outcome 1 with probability $|\beta|^2$

After the measurement, the state collapses to the corresponding eigenstate.

This is the quantum analogue of reading a classical bit.

ii. Measurement in a different basis

A qubit can also be measured in other bases, such as:

σ_x basis:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad , \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

→ Probabilities →

$$P(+)=|\langle+|\psi\rangle|^2 \quad , \quad P(-)=|\langle-|\psi\rangle|^2$$

σ_y basis:

$$|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}} \quad , \quad |-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}.$$

→ Probabilities →

$$P(+i)=|\langle+i|\psi\rangle|^2 \quad , \quad P(-i)=|\langle-i|\psi\rangle|^2$$

iii. What does “measuring in a different basis” mean physically?

It does not mean the system changes.

It means we choose a different question to ask the same system.

- Measurement in the σ_z basis asks:

“Is the qubit in $|0\rangle$ or $|1\rangle$?”

- Measurement in the σ_x basis asks:

“Is the qubit aligned or anti-aligned with the x-axis?”

- Measurement in the σ_y basis asks:

“Is the qubit aligned or anti-aligned with the y-axis?”

The system is the same. The measurement apparatus is oriented differently.

This is exactly like measuring the spin of an electron along different axes.

iv. Why do different bases give different probabilities?

Because the qubit state has a relative phase and a population distribution that depend on the basis.

Example:

$$|\psi_1\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad ; \quad |\psi_2\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

In the σ_z basis:

$P(0) = P(1) = 1/2$, for both states

But in the σ_x basis:

$$|\psi_1\rangle = |+\rangle \quad , \quad |\psi_2\rangle = |-\rangle$$

So

- ψ_1 gives outcome + with probability 1
- ψ_2 gives outcome – with probability 1

The difference comes entirely from the relative phase.

This is the essence of quantum interference.

Why a Qubit Is Not a Classical Bit

A classical bit has only two states: 0 or 1.

A qubit has:

- infinitely many pure states
- continuous parameters θ & ϕ
- basis-dependent measurement outcomes
- non-commuting observables
- interference effects

- collapse upon measurement

This richer structure is what gives quantum computing its power.